TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Passive damping technology, e.g. to increase the audio performance in speakers and car audio			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical Electronics		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Imaging/Audio Devices	Speakers	Precision Manufacturing Machinery	Cutting Machine Tools
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* To increase the performance of a speaker and in car audio systems, parasitic oscillations in every part of the acoustic systems, especially in the speaker chassis and speaker cabinet, have to be reduced. The best way is to integrate stiff high-damping materials that are able to convert the energy of unwanted oscillations into heat. The high damping composite material developed at Fraunhofer IFAM Dresden is the best choice for this purpose. The sound after integration will be clearer, provide more amplitude power for each frequency and reduce parasitic oscillations to a minimum.			
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* The origin of parasitic oscillations varies for different applications and designs. While the basic concept of the (composite) materials exists, material systems have to be adopted for different			

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	applications, balancing the requirements for a limited number of specific alloys and the individual circumstances for each application. This needs to be addressed in a joint R&D project regarding specific applications, products and new developments in combination with the possible material specifications and the specific use-case requirements from the industrial partner. Once the strategy of integration is agreed and verified by simulations, the unique material can be integrated in a prototype phase. After evaluation of the system demonstrator with the equipment of the Korean industrial partner next collaboration steps can be discussed to up-scale to industrial level.
Key Words	1. *Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 2. Reduction of unwanted parasitic oscillations 3. High damping material with high stiffness and damping 4. Audio application 5. Noise reduction 6. Efficiency increase

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* The main goal is to develop a metal graphite composite materials and its production process that allows for a significant improvement of sound quality in a car audio application.			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* - Increase of amplitude of generated frequency - Faster reduction of generated noise after switching off the signal generation - Reduction of parasitic oscillations (up to 10dB) that can overlay the wanted signal - subjectively better acoustic perception by the listener - Passive solution that can be integrated in speaker and car acoustic applications			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* Phase 1 (6 months): Evaluation of actual design and acoustic performance Korea: Produce actual design of speaker and/ or car acoustic system with production evaluation Germany: Acoustic evaluation and development of possible integration opportunities of high damping material into actual design Phase 2.1 (18 months): Integration of high damping material and evaluation in actual design Korea: Provide actual design, parts, components for integration process of high damping material; Acoustic evaluation of integrated version and comparison with original setup. Germany: Development and manufacturing of high damping material; Integration into actual design; Acoustic evaluation of integrated version compared with original setup Phase 2.2 (12 months): Integration of high damping material and evaluation of new design and build up of prototype Korea: Development and manufacturing of new design with integrated high damping material for acoustic evaluation Germany: Manufacturing of high damping material and integration in new design; acoustic evaluation Milestone (month 24 of phase 2): Integration and acoustic evaluation document better acoustic performance of the integrated setup compared to the original setup			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* - Development and manufacturing of high damping material with powder metallurgical methods - Integration of high damping material into actual and new design - Acoustic evaluation of actual and new design - Material characterization regarding stiffness, damping, density
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* - Provide actual design of speaker and car acoustic system - Perform acoustic evaluation - Provide, manufacture parts and components to build up a whole system with and without the integration of high damping material - Develop new designs with integrated high damping material - Manufacturing new design as prototype

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* - After integration of high damping material the speaker and car acoustic system will perform better in terms of clearness, frequency precision and reduction of parasitic oscillations - Including the high damping material in the design phase of new products will generate an unique selling point for the industry - Integration in the right place can boost acoustic comfort of the listeners - Increase market penetration opportunities through increased acoustic performance - Plenty of additional damping applications, e.g. in precision manufacturing

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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